

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

*I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3. Given the concept map:

DataLink → MACAddress → LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→ DataLinkLayer → ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping

Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

1. Complete the OSI Concept Map

OSI Model:

Application → Presentation → _____ → Transport → Network → _____ → Physical

Missing Layers: Session Layer, Data Link Layer

Functions:

- **Session Layer:** Establishes, manages, synchronizes, and terminates communication sessions between devices.
- **Data Link Layer:** Organizes data into frames, provides MAC addressing, detects errors, and controls data flow for reliable communication.

This follows the same style as the example in your image, with the concept map first, followed by the missing layers and their functions.

2. Complete the Data Units and OSI Layers Concept Map

Application / Presentation / Session → _____ → Network → _____ → Physical

Data Units:

Data → _____ → Packet → _____ → Bits

Missing Terms: Segment, Frame

Completed Concept Map:

- Application / Presentation / Session → Data
- Transport → Segment
- Network → Packet
- Data Link → Frame
- Physical → Bits

Analysis:

- **Data:** Created by the Application, Presentation, and Session layers.
- **Segment:** The Transport layer divides data into segments for reliable end-to-end delivery.
- **Packet:** The Network layer adds logical addressing (IP) to create packets for routing.
- **Frame:** The Data Link layer encapsulates packets into frames with MAC addresses and error detection.
- **Bits:** The Physical layer converts frames into bits for transmission over the communication medium.

This transformation of **Data** → **Segment** → **Packet** → **Frame** → **Bits** ensures that information is properly addressed, checked for errors, and transmitted reliably from the sender to the receiver.

3. Complete the Networking Concept Map

Data Link → _____ → Local Delivery

Network → _____ → Global Delivery

Missing Terms: MAC Address, IP Address

Completed Concept Map:

- Data Link → MAC Address → Local Delivery
- Network → IP Address → Global Delivery

Analysis:

- **MAC Address:** Used by the **Data Link layer** to identify devices within the same local network (LAN), ensuring data reaches the correct device.
- **IP Address:** Used by the **Network layer** to identify devices across different networks, enabling data to be routed globally.
- **Together:** The **MAC Address** ensures correct delivery within the local network, while the **IP Address** enables communication between different networks, ensuring complete end-to-end data delivery.

4. Complete the Error Control Concept Map

Error Control

↓

Data Link Layer → _____

↓

Transport Layer → _____

Missing Terms: Error Detection, Error Correction

Completed Concept Map:

- Error Control → Data Link Layer → Error Detection
- Error Control → Transport Layer → Error Correction

Analysis:

- **Data Link Layer:** Detects errors in frames during transmission over the local network using techniques such as error-checking codes.
- **Transport Layer:** Corrects transmission errors by requesting retransmission of lost or corrupted data and ensuring reliable end-to-end delivery.
- **Together:** The Data Link layer identifies transmission errors, while the Transport layer recovers from those errors. Their interaction improves the accuracy, reliability, and integrity of data transmission across the network.

5. Develop the Web Browsing OSI Concept Map

Web Browser → Application → Presentation → Session → Transport → Network → Data Link → Physical → Web Server

Analysis:

- **Application Layer:** Creates and receives web requests (HTTP/HTTPS).
- **Presentation Layer:** Encrypts, decrypts, and formats data (SSL/TLS).
- **Session Layer:** Establishes and manages the communication session.
- **Transport Layer:** Ensures reliable delivery of data using TCP.
- **Network Layer:** Routes packets using IP addresses.
- **Data Link Layer:** Delivers frames within the local network using MAC addresses.
- **Physical Layer:** Transmits data as bits through cables, Wi-Fi, or fiber.

Effect of Layer Failure:

- **Application Layer Failure:** Web pages cannot be requested or displayed.
- **Presentation Layer Failure:** Data cannot be encrypted/decrypted or interpreted correctly.
- **Session Layer Failure:** The communication session is interrupted or disconnected.
- **Transport Layer Failure:** Data may be lost, duplicated, or arrive out of order.
- **Network Layer Failure:** Packets cannot reach the destination due to routing issues.
- **Data Link Layer Failure:** Frames may be corrupted or not delivered on the local network.
- **Physical Layer Failure:** No bits are transmitted, causing complete communication failure.

6. Analyze the Network Troubleshooting Concept Map

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analysis:

- **Physical Layer ✓:** The physical connection (cables, Wi-Fi, signals) is working correctly.
- **Data Link Layer ✓:** Frames are successfully transmitted within the local network.
- **Network Layer ✗:** Routing or IP addressing has failed, so packets cannot reach the destination.
- **Transport Layer ✗:** Since the Network layer has failed, reliable end-to-end communication cannot be established.
- **Application Layer ✗:** Applications (such as web browsers or email) cannot communicate because the lower layers are not delivering data.

How the Concept Map Helps in Troubleshooting:

- **✓ Physical Layer** → Confirms the hardware connection is functioning.
- **✓ Data Link Layer** → Confirms local network communication is working.
- **✗ Network Layer** → Identifies the main problem (IP addressing or routing).
- **✗ Transport Layer** → Fails because packets cannot travel through the network.
- **✗ Application Layer** → Fails because no data reaches the application.